

Answer ALL questions.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

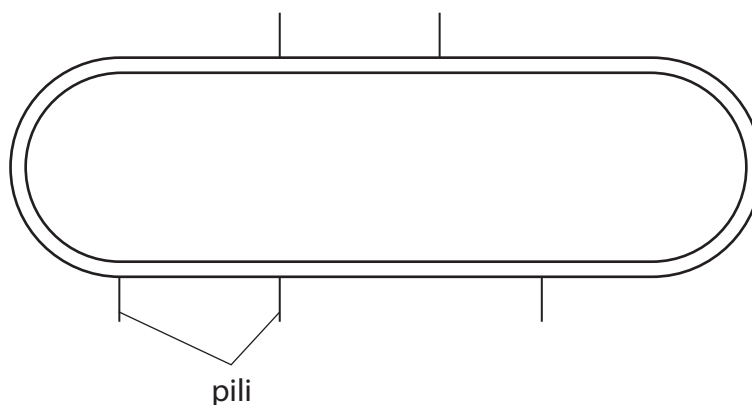
1 Woese classified organisms into a system with three domains.

(a) How many of the domains contain prokaryotic organisms?

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

(b) The diagram shows part of a prokaryotic cell, as drawn by a student.



(i) Complete the diagram by drawing and labelling a flagellum and a plasmid.

(2)

(ii) State the function of a plasmid.

(1)

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- (c) (i) The ribosomes in prokaryotic cells are different from the ribosomes in eukaryotic cells.

Give **one** difference between prokaryotic ribosomes and eukaryotic ribosomes.

(1)

- (ii) How many of the following statements about eukaryotic ribosomes are correct?

- ribosomes are located on the surface of both types of endoplasmic reticulum
- ribosomes are located inside chloroplasts and mitochondria
- ribosomes are involved in the process of transcription

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

(Total for Question 1 = 6 marks)